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B.M.S. College of Engineering, Bengaluru-560019

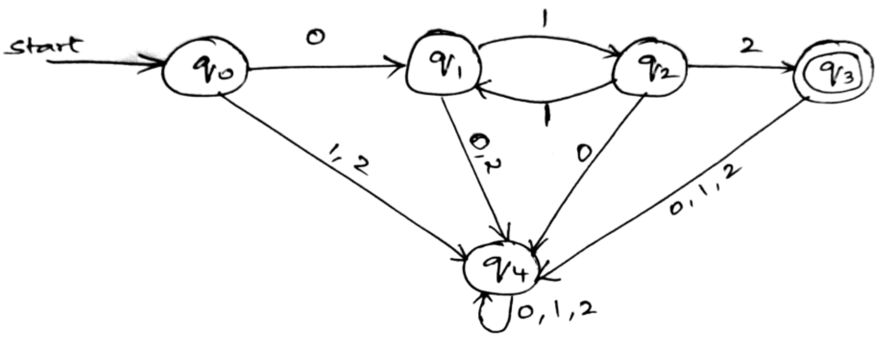
Autonomous Institute Affiliated to VTU

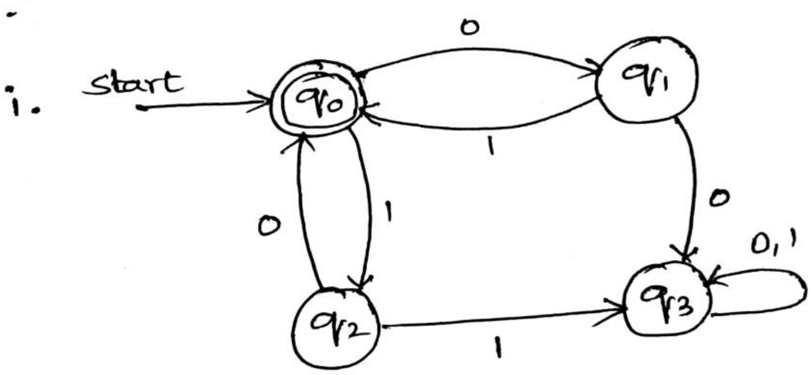
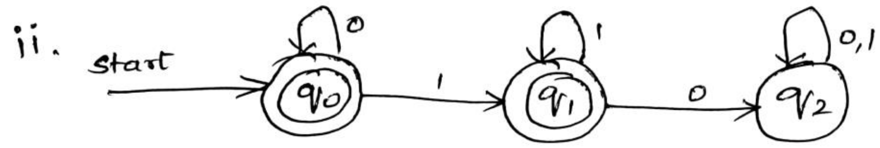
February 2025 Semester End Main Examinations

Programme: B.E. | **Semester: IV**
Branch: Computer Science and Engineering | **Duration: 3 hrs.**
Course Code: 22CS4PCTFC | **Max Marks: 100**
Course: Theoretical Foundations of Computations

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	*CO*	*PO*	**Marks**
1	a)	Define Finite Automata. Demonstrate the same using an ON/OFF Switch.	*1*	*1*	**5**
	b)	Analyze the given DFA. Draw the transition table and write the language equivalent. Explain.  	*1*	*2*	**7**
	c)	Obtain an ε-NFA which accepts strings consisting of zero or more **a**'s followed by zero or more **b**'s followed by zero or more **c**'s.	*1*	*3*	**8**
		OR			
2	a)	Design DFA for accepting binary strings that i)start with 00 ii)having 001 iii) ending with 01 iv) not having 111 v)whose value is multiple of 2	*3*	*3*	**10**
	b)	Design NFA for accepting strings that have 1 in second last position. Convert the same to DFA.	*1, 3*	*1,3*	**10**
		UNIT - II			
3	a)	List the notations used to represent regular expressions. Demonstrate the same with an example.	*1*	*1*	**6**

	b)	Analyze the given Finite Automata (FA) and obtain the corresponding regular expression.  	*1*	*2*	**6**
	c)	Obtain an Non- Deterministic Finite Automata (NFA) for the regular expression (a+b)*aa(a+b)*. Show the step by step procedure involved in constructing the NFA.	*1*	*3*	**8**
		OR			
4	a)	Explain closure property – complementation with an example.	*1*	*1*	**6**
	b)	Write the regular expressions for the following. i. L={w : n_a(w)mod 3 =0 where w=(a,b)*} ii. L={a^nb^m | m ≥ 1, n ≥ 1, nm ≥ 3}	*1*	*1*	**6**
	c)	Obtain the distinguishable table for the automaton and then minimize the states of the following DFA. | δ | a | b | | →**A** | B | F | | **B** | G | C | | ***C** | A | C | | **D** | C | G | | **E** | H | F | | **F** | C | G | | **G** | G | E | | **H** | G | C |	*1*	*3*	**8**
		UNIT - III			
5	a)	Show that the below grammar is ambiguous. S → iCtS | iCtSeS | a , C → b	*1*	*1*	**4**
	b)	Consider the following context-free grammars. G1: S-> aS | B, B-> b | bB	*1*	*2*	**8**

			G2: S->aA | bB, A-> aA | B | ε , B-> bB | ε Write the Context free language generated by G1 and G2			
		c)	Analyze the given language and obtain a grammar to generate the following languages: i. $L=\{a^{n+2} b^m \mid n\geq 0 \text{ and } m>n\}$ ii. $L=\{a^n b^m c^k \mid n\geq 0 \text{ and } m>n\}$	*2*	*2*	**8**
			OR			
	6	a)	Eliminate useless symbols in the grammar given below: {S →aA | bB , A → aA | a , B →bB , D →ab | Ea , E →aC | d}	*3*	*1*	**10**
		c)	Consider the grammar. S -> 0A | 1B A -> 0AA | 1S | 1 B -> 1BB | 0S | 0 Obtain the grammar in Chomsky Normal Form (CNF).	*1*	*3*	**10**
			UNIT - IV			
	7	a)	State the pumping lemma for Context Free languages. Explain with an example.	*2*	*2*	**5**
		b)	Check if the Push Down Automata (PDA) to accept the language $L(M)=\{w \mid w\ (a+b)^* \text{ and } n_a(w) = n_b(w)$ is deterministic or non-deterministic?	*3*	*3*	**5**
		c)	For the given grammar, obtain the corresponding PDA. S -> aABB | aAA A -> aBB | a B -> bBB | A C -> a	*1*	*1*	**10**
			OR			
	8	a)	Design PDA for accepting $a^n b^{2n}$ such that n>=1.	*3*	*3*	**10**
		b)	Design PDA to accept $a^n b^n$. Also check whether the PDA is DPDA or NPDA.	*3*	*3*	**10**
			UNIT - V			
	9	a)	Discuss about the various components of Turing machine model.	*3*	*3*	**5**
		b)	Find whether the lists given here have a Post Correspondence Solution. M = (abb, aa, aaa) and N = (bba, aaa, aa)	*3*	*3*	**5**
		c)	Design a Turing machine to accept the language $L(M)=\{0^n 1^n 2^n \mid n \geq 1\}$	*3*	*3*	**10**
			OR			

	10	a)	Design TM that adds two numbers. Assume numbers are stored on tape as sequence of 1's and separated by a 0.	*3*	*3*	**10**
		b)	Design TM that accepts palindrome strings of a and b.	*3*	*3*	**10**

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

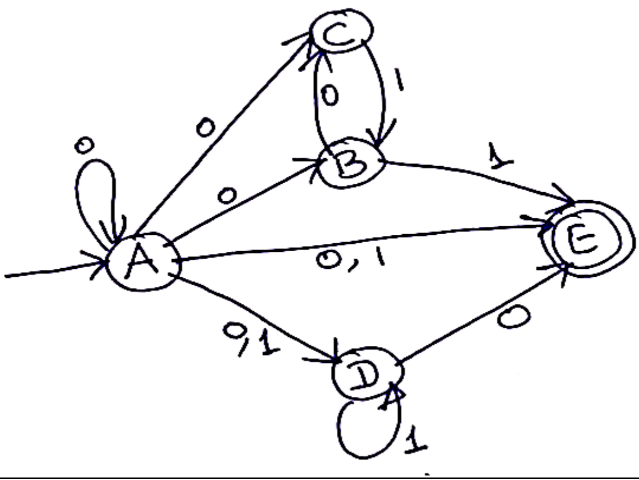
August 2024 Semester End Main Examinations

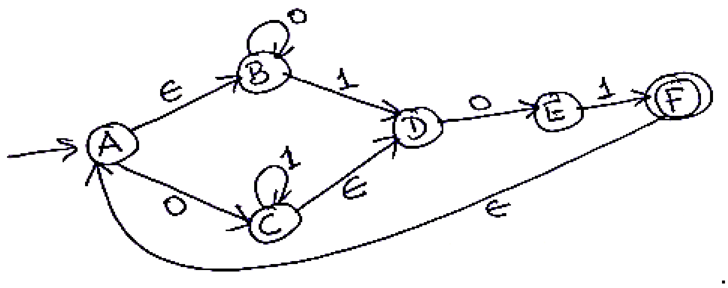
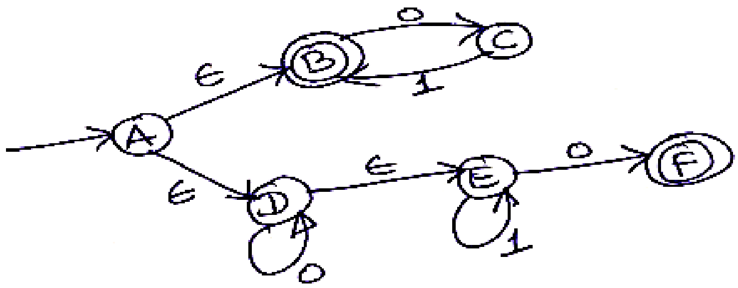
Programme: B.E.
Branch: Computer Science and Engineering
Course Code: 23CS4PCTFC
Course: Theoretical Foundations of Computations

Semester: IV
Duration: 3 hrs.
Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	***CO***	***PO***	**Marks**
1	a)	Define Deterministic Finite Automata (DFA). Construct the DFA to accept the language L= {w | w ∈{a, b}* and w ends with abb}. Show the DFA transition function movements for the string abaabb	***CO3***	***PO 3***	**10**
	b)	For the Non-deterministic Finite Automata (NFA) shown below, using the subset construction method to find the equivalent DFA. 	***CO3***	***PO3***	**10**
		OR			
2	a)	Define NFA. Construct the NFA to accept the language L= {w | w ∈ {a, b}* and w has exactly two a’s}. Show the NFA transition function movements for the string babbab	***CO3***	***PO3***	**6**
	b)	Define ε-closure (A), where A is any state of the given ε-NFA (Epsilon NFA). Compute the ε-closure of the states {A, B, C, F} from the following ε-NFA.	***CO1***	***PO1***	**4**

						
		c)	Convert following ε-NFA (Epsilon NFA) to DFA using Epsilon (ε) closure. 	*CO3*	*PO3*	**10**
			UNIT - II			
	3	a)	Design Regular Expressions (RE) which generates the following languages over the alphabet set $\sum = \{0, 1\}$. i. Set of all strings ending with 1 and not containing 00. ii. Set of all strings that do not contain the substring 01.	*CO3*	*PO3*	**6**
		b)	Show that the regular languages are closed under Difference and Intersection operation.	*CO2*	*PO2*	**6**
		c)	State pumping lemma for regular languages. Use pumping lemma to show that the following language is not regular. $L = \{xx^R \mid x \in (0, 1)^*\}$.	*CO2*	*PO2*	**8**
			UNIT - III			
	4	a)	Design Context Free Grammar (CFG) to generate each of the following languages. i. $L=\{a^ib^jc^k \mid j=i+k\}$ ii. $L=\{ a^ib^jc^k \mid j=i \text{ or } j=k\}$	*CO2*	*PO2*	**6**
		b)	Let G be the CFG with productions set as $\{S \rightarrow S+S \mid S-S \mid S*S \mid S/S \mid (S) \mid a\}$. Answer the following w.r.t a+(a*a)/a-a. i. Give two left most derivations. ii. Draw the derivation tree corresponds to each of the derivations in (i). iii. How many distinct leftmost derivations are there?	*CO3*	*PO3*	**6**

		c)	Convert following CFG to Greibach Normal Form (GNF). $S \rightarrow AaA \mid CA \mid BaB$ $A \rightarrow aaBa \mid CDA \mid aa \mid DC$ $B \rightarrow bB \mid bAB \mid bb \mid aS$ $C \rightarrow Ca \mid bC \mid D$ $D \rightarrow bD \mid \varepsilon$	*CO2*	*PO2*	**8**
			OR			
	5	a)	Write CFG to generate each of the following languages. i. $L=\{a^ib^jc^k \mid i < j\}$ ii. $L=\{a^ib^jc^k \mid j < k\}$	*CO2*	*PO2*	**6**
		b)	Answer the following i. Identify the language generated by the CFG with productions $S \rightarrow aSaSbS \mid aSbSaS \mid bSaSaS \mid \varepsilon$ ii. Show that the CFG with productions $S \rightarrow aSb \mid aaSb \mid \varepsilon$ is ambiguous.	*CO2*	*PO2*	**6**
		c)	Define Chomsky Normal Form (CNF). Convert following Grammar to CNF. $S \rightarrow ABA$, $A \rightarrow aA \mid \varepsilon$, $B \rightarrow bB \mid \varepsilon$	*CO3*	*PO3*	**8**
			UNIT – IV			
	6	a)	Define Push Down Automata (PDA). Construct Nondeterministic Push Down Automata (NPDA) for the language $L=\{w \mid w \in (a,b)^*$ and w is palindrome of even length$\}$ by final state method. Show the Instantaneous Description (ID) for the string abaaba is accepted by the NPDA constructed.	*CO3*	*PO3*	**8**
		b)	Convert following Grammar to PDA. Show the instantaneous description for the string "aabba" $S \rightarrow aABC$ $A \rightarrow aB \mid a$ $B \rightarrow bA \mid b$ $C \rightarrow a$	*CO2*	*PO2*	**6**
		c)	State pumping lemma for Context Free Languages (CFL). Apply pumping lemma to show the language $L=\{a^ib^jc^k \mid i< j<k\}$ is not a context free language.	*CO2*	*PO2*	**6**
			UNIT - V			
	7	a)	Design the Turing Machine (TM) which accepts the set of all palindromes over the alphabet set $\Sigma = \{0, 1\}$. Trace the operation of the constructed TM on the string 100001	*CO3*	*PO3*	**10**
		b)	Define Post Correspondence Problem (PCP). Find a Post Correspondence Solution for following two lists given.	*CO3*	*PO3*	**6**

			<table><tr><td></td><td>List X</td><td>List Y</td></tr><tr><td>i</td><td>Xi</td><td>Yi</td></tr><tr><td>1</td><td>10</td><td>101</td></tr><tr><td>2</td><td>01</td><td>100</td></tr><tr><td>3</td><td>0</td><td>10</td></tr><tr><td>4</td><td>100</td><td>0</td></tr><tr><td>5</td><td>1</td><td>010</td></tr></table>			
		c)	Describe Multitape Turing Machine.	***CO1***	***PO1***	**4**

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B.M.S. College of Engineering, Bengaluru-560019

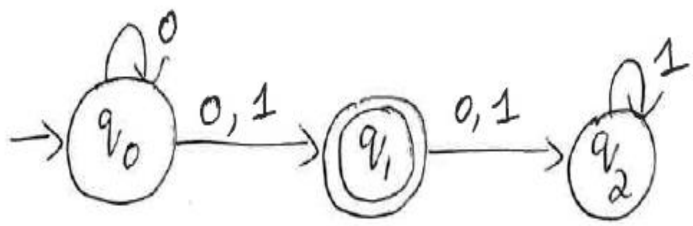
Autonomous Institute Affiliated to VTU

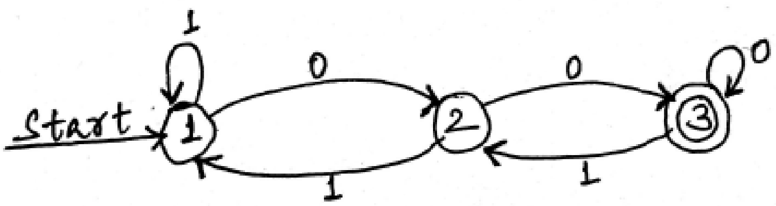
August 2024 Semester End Main Examinations

Programme: B.E. | **Semester: IV**
Branch: Computer Science & Engineering | **Duration: 3 hrs.**
Course Code: 22CS4PCTFC | **Max Marks: 100**
Course: Theoretical Foundations of Computations

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	***CO***	***PO***	**Marks**
1	a)	Construct a Deterministic Finite Automata (DFA) to accept a string of a's and b's having not more than three a's	*CO3*	*PO3*	**5**
	b)	List the applications of Automata and explain	*CO1*	*PO2*	**5**
	c)	Convert the given Non-Deterministic Finite Automata (NFA) to its equivalent DFA 	*CO2*	*PO2*	**10**
		UNIT - II			
2	a)	Obtain regular expression to accept the strings of a's and b's such that every block of four consecutive symbols containing at least two **b**'s.	*CO1*	*PO1*	**5**
	b)	Using Pumping Lemma, show that the language $\mathbf{L = \{a^{n!} \mid n>=0\}}$ is not regular. Note: n! represents n factorial.	*CO2*	*PO2*	**6**

<table>
<tr><td></td><td></td><td>c)</td><td>For the given transition table, obtain the minimized DFA for the following:
<table>
<tr><th>δ</th><th>0</th><th>1</th></tr>
<tr><td>→ A</td><td>B</td><td>A</td></tr>
<tr><td>B</td><td>A</td><td>C</td></tr>
<tr><td>C</td><td>D</td><td>B</td></tr>
<tr><td>*D</td><td>D</td><td>A</td></tr>
<tr><td>E</td><td>D</td><td>F</td></tr>
<tr><td>F</td><td>G</td><td>E</td></tr>
<tr><td>G</td><td>F</td><td>G</td></tr>
<tr><td>H</td><td>G</td><td>D</td></tr>
</table></td><td>CO1</td><td>PO1</td><td>9</td></tr>
<tr><td></td><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>3</td><td>a)</td><td>Obtain Regular Expressions (RE) for the following languages over the alphabet set Σ ={a,b}
i. Set of string of a's and b's having substring "aa"
ii. Strings of a's and b's whose lengths are multiples of 3
iii. Strings that do not end with ab</td><td>CO1</td><td>PO1</td><td>5</td></tr>
<tr><td></td><td></td><td>b)</td><td>Analyze the following NFA and obtain the regular expression using Kleene's technique for the following DFA.

</td><td>CO2</td><td>PO2</td><td>10</td></tr>
<tr><td></td><td></td><td>c)</td><td>Show that L= {a^ib^j |i>j} is NOT regular using pumping lemma</td><td>CO2</td><td>PO2</td><td>5</td></tr>
<tr><td></td><td></td><td></td><td>UNIT - III</td><td></td><td></td><td></td></tr>
<tr><td></td><td>4</td><td>a)</td><td>Obtain a grammar to generate a language consisting of all non-palindromes over {a,b}. Give the derivation for the string ababba which is not a palindrome.</td><td>CO2</td><td>PO2</td><td>6</td></tr>
<tr><td></td><td></td><td>b)</td><td>Show that the following grammar is ambiguous.

S -> AB | aaB
A -> a | Aa
B -> b</td><td>CO2</td><td>PO2</td><td>4</td></tr>
<tr><td></td><td></td><td>c)</td><td>Covert the following grammar to its corresponding Chomsky Normal Form (CNF).

S -> 0A | 1B
A -> 0AA | 1S | 1
B -> 1BB | 0S | 0</td><td>CO2</td><td>PO2</td><td>10</td></tr>
</table>

			OR			
	5	a)	Show that the following grammar is ambiguous over **w**= (( ) ( ) ( )) S -> SS | (S) | ε	*CO2*	*PO2*	**5**
		b)	Eliminate useless symbols for the following grammar S -> aA | a | Bb | cC A -> aB B -> a | Aa C -> cCD D -> ddd	*CO2*	*PO2*	**8**
		c)	Analyze the following grammar and eliminate all unit productions from the grammar. S -> Aa | B | Ca B -> aB | b C -> Db | D D -> E | d E -> ab	*CO2*	*PO2*	**7**
			UNIT - IV			
	6	a)	Design Push Down Automata (PDA) for the language L={$a^n b^{2n}$ | n>=1}. Write an Instantaneous Description for the string **aabbbb**	*CO3*	*PO3*	**5**
		b)	Convert the following grammar to PDA S -> aABB | aAA A -> aBB | a B -> bBB | b C -> a	*CO2*	*PO2*	**10**
		c)	Show Context Free Languages (CFL) are not closed under Intersection	*CO2*	*PO2*	**5**
			UNIT - V			
	7	a)	Differentiate between Recursive Language and Recursively Enumerable Languages	*CO2*	*PO2*	**5**
		b)	Obtain a Turing Machine which accepts the language L={$0^n 1^n$ | n>=1}	*CO3*	*PO3*	**10**
		c)	Describe undecidable problems	*CO1*	*PO1*	**5**

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

September / October 2024 Supplementary Examinations

Programme: B.E. **Semester: IV**
Branch: Computer Science and Engineering **Duration: 3 hrs.**
Course Code: 22CS4PCTFC **Max Marks: 100**
Course: Theoretical Foundations of Computations

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
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<table>
<tr><th></th><th></th><th>UNIT - I</th><th>CO</th><th>PO</th><th>Marks</th></tr>
<tr><td>1</td><td>a)</td><td>Construct Deterministic Finite Automata (DFA) accepting the following strings over the alphabet set $\Sigma = \{0,1\}$
i. The set of all strings ending with 00
ii. The set of all strings with three consecutive 0’s atleast once
iii. Set of all strings beginning with 00</td><td>CO3</td><td>PO 3</td><td>10</td></tr>
<tr><td></td><td>b)</td><td>Convert the following Non- Deterministic Finite Automata (NFA) to a DFA
<table>
<tr><th></th><th>0</th><th>1</th></tr>
<tr><td>$\rightarrow p$</td><td>$\{q, s\}$</td><td>$\{q\}$</td></tr>
<tr><td>$*q$</td><td>$\{r\}$</td><td>$\{q, r\}$</td></tr>
<tr><td>r</td><td>$\{s\}$</td><td>$\{p\}$</td></tr>
<tr><td>$*s$</td><td>$\emptyset$</td><td>$\{p\}$</td></tr>
</table></td><td>CO2</td><td>PO2</td><td>5</td></tr>
<tr><td></td><td>c)</td><td>Construct an NFA over {0, 1} that accepts the set of strings that contain an even number of substrings 01</td><td>CO3</td><td>PO3</td><td>5</td></tr>
<tr><td></td><td></td><td>UNIT - II</td><td></td><td></td><td></td></tr>
<tr><td>2</td><td>a)</td><td>Design Regular Expressions (RE) for the following language over $\Sigma = \{a,b\}$
i. Set of string of a’s and b’s ending with string “abb”
ii. Set of string of a’s and b’s starting with string “ab”
iii. Set of string of a’s and b’s having substring “aa”</td><td>CO3</td><td>PO3</td><td>10</td></tr>
</table>

<table>
<tr><td></td><td></td><td></td><td>iv. Set of strings consisting of atleast one “a” followed by string consisting of atleast one “b”
v. Set of strings that contain exactly three a’s</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>b)</td><td>Convert the following DFA to a RE, using the Kleene’s technique:
<table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→p</td><td>q</td><td>p</td></tr><tr><td>*q</td><td>q</td><td>q</td></tr></table></td><td>CO3</td><td>PO3</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>3</td><td>a)</td><td>Design minimized DFA using the concept of table filling algorithm for the DFA given below.
<table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→ A</td><td>B</td><td>E</td></tr><tr><td>B</td><td>C</td><td>F</td></tr><tr><td>*C</td><td>D</td><td>H</td></tr><tr><td>D</td><td>E</td><td>H</td></tr><tr><td>E</td><td>F</td><td>I</td></tr><tr><td>*F</td><td>G</td><td>B</td></tr><tr><td>G</td><td>H</td><td>B</td></tr><tr><td>H</td><td>I</td><td>C</td></tr><tr><td>*I</td><td>A</td><td>E</td></tr></table></td><td>CO3</td><td>PO3</td><td>10</td></tr>
<tr><td></td><td></td><td>b)</td><td>Using pumping lemma show that the language
$L = \{a^n b^n, n >= 0\}$ is not regular.</td><td>CO2</td><td>PO2</td><td>5</td></tr>
<tr><td></td><td></td><td>c)</td><td>Show that regular languages are closed under Intersection with an example.</td><td>CO1</td><td>PO1</td><td>5</td></tr>
<tr><td></td><td></td><td></td><td>UNIT - III</td><td></td><td></td><td></td></tr>
<tr><td></td><td>4</td><td>a)</td><td>Design Context Free Grammar (CFG) to accept following languages
i. $L=\{a^n b^n, n>=0\}$
ii. $L=\{a^{n+1} b^n, n>=0\}$
iii. $L=\{a^n b^{2n}, n>=0\}$
iv. $L=\{w, |w| \bmod 3 > 0 , \text{ where } w \in a^*\}$</td><td>CO3</td><td>PO3</td><td>10</td></tr>
<tr><td></td><td></td><td>b)</td><td>Convert below grammar to Chomsky Normal Form (CNF).
$S \rightarrow aXbX$, $X \rightarrow aY \mid bY \mid \boldsymbol{\varepsilon}$, $Y \rightarrow X \mid c$</td><td>CO2</td><td>PO2</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>5</td><td>a)</td><td>Eliminate useless symbols in the grammar given below:
$\{S \rightarrow aA \mid bB , A \rightarrow aA \mid a , B \rightarrow bB , D \rightarrow ab \mid Ea , E \rightarrow aC \mid d\}$</td><td>CO2</td><td>PO2</td><td>10</td></tr>
<tr><td></td><td></td><td>b)</td><td>Eliminate unit productions in the below grammar
$S \rightarrow AB$
$A \rightarrow a$
$B \rightarrow C \mid b$</td><td>CO2</td><td>PO2</td><td>5</td></tr>
</table>

			$C \rightarrow D$ $D \rightarrow E \mid bC$ $E \rightarrow d \mid Ab$			
	c)	Show that the below grammar is ambiguous. $S \rightarrow iCtS \mid iCtSeS \mid a$, $C \rightarrow b$	*CO1*	*PO1*	**5**
		UNIT - IV			
6	a)	Design Push Down Automata (PDA) for the language $L=\{WcW^R \mid W$ is a string of a's and b's$\}$. Show the Instantaneous Description for the string "aabcbaa"	*CO3*	*PO3*	**10**
	b)	Obtain CFG that generates the language accepted by the PDA given below $P=(Q, \sum, \Gamma, q_0, Z_0, \delta, F)$ $Q = \{ q_0, q_1, q_2 \}$ $\sum = \{a, b\}$ $\Gamma = \{A, Z\}$ $q_0 = q_0$ $Z_0 = Z$ $F = \{q_2\}$ $\delta(q_0, a, Z)=(q_0, AZ)$ $\delta(q_0, a, A)=(q_0, A)$ $\delta(q_0, b, A)=(q_1, \varepsilon)$ $\delta(q_1, \varepsilon, Z)=(q_2, \varepsilon)$	*CO2*	*PO2*	**10**
		UNIT - V			
7	a)	Design Turing Machine to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0) Give the formal definition of the obtained TM.	*CO3*	*PO3*	**10**
	b)	Describe "Undecidable" problem with an example.	*CO1*	*PO1*	**5**
	c)	Describe Multitape Turning Machine.	*CO1*	*PO1*	**5**

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B.M.S. College of Engineering, Bengaluru-560019

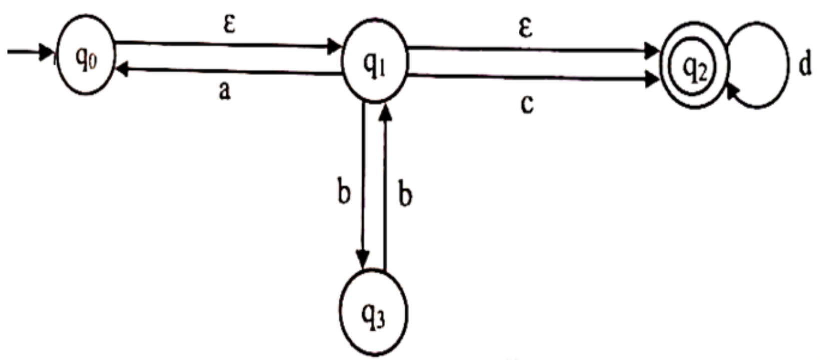
Autonomous Institute Affiliated to VTU

October 2024 Supplementary Examinations

Programme: B.E. **Semester: IV**
Branch: Computer Science and Engineering **Duration: 3 hrs.**
Course Code: 23CS4PCTFC **Max Marks: 100**
Course: Theoretical Foundations of Computations

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	CO	PO	Marks
1	a)	(i) Design Deterministic Finite Automata (DFA) which accepts set of strings such that every string containing 00 as a substring but not 000 as substring. Consider alphabet set as Σ ={0, 1}. (ii) Design NFA for the language L= {all strings over (0,1)* that have at least two consecutive 0's or two consecutive 1's }	*CO1*	*PO1*	**5**
	b)	Convert the following Non-Deterministic Finite Automata (NFA) with Epsilon (ε)-transitions to equivalent NFA without Epsilon - transitions by giving the epsilon closure of all the states. 	*CO 1*	*PO 1*	**8**
	c)	Construct NFA which accepts set of all strings whose second last symbol is 1. Convert this NFA to DFA using Subset Construction Method. Consider alphabet set as Σ ={0, 1}.	*CO 1*	*PO 1*	**7**
		OR			
2	a)	(i) Design DFA to accept strings of a's and b's ending with ab or ba (ii) Design a DFA to accept the language L={w | number of a's in (w) >=1, number of b's in (w) >=2}	*CO 1*	*PO 1*	**5**

b) Convert the following NFA to DFA using Subset Construction Method. — *CO 1* — *PO 1* — **8**

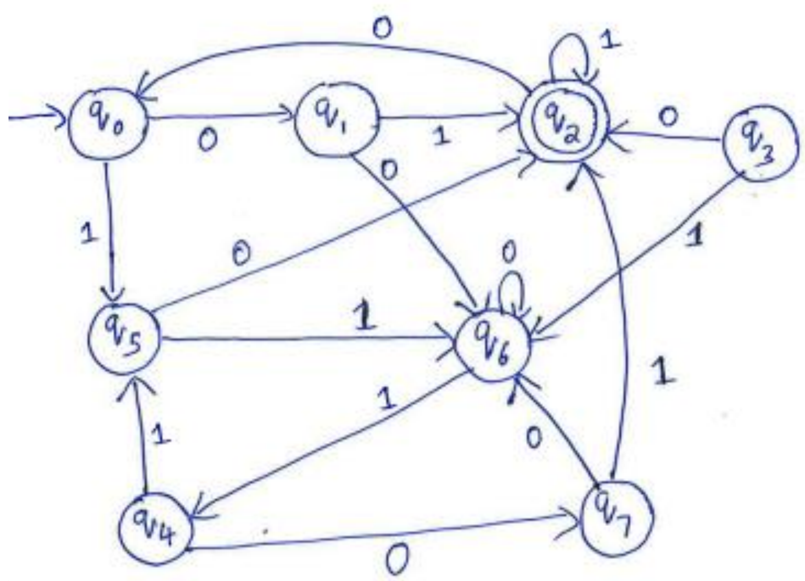
δ	0	1
→ Q0	Q0, Q1	Q1
* Q1	Q2	Q2
Q2	∅	Q2

c) Convert the following ε-NFA to DFA. — *CO 1* — *PO 1* — **7**

δ	ε	a	b	c
→ P	∅	P	Q	R
Q	P	Q	R	∅
* R	Q	R	∅	P

UNIT-II

3 a) Construct the minimum state automata for the finite automata given below. Consider alphabet set as $\Sigma = \{0, 1\}$. — *CO 2* — *PO 2* — **10**



b) Convert the DFA which accepts Even number of 1's to Regular Expression (RE) by eliminating intermediate states method. Consider alphabet set as $\Sigma = \{0, 1\}$. — *CO 2* — *PO 2* — **10**

UNIT - III

4 a) Convert the following Grammar to Chomsky Normal Form (CNF) — *CO 2* — *PO 2* — **10**

$S \rightarrow abAB$, $A \rightarrow Bab \mid \varepsilon$, $B \rightarrow Baa \mid A \mid \varepsilon$

b) Design a Context Free Grammar (CFG) for $L = \{ a^n c^m b^k \mid n=m \text{ or } m<=k ,\ \text{for } n, m, k >=0 \}$. Write formal definition of the obtained CFG. — *CO 2* — *PO 2* — **10**

			OR			
	5	a)	Convert the following Grammar to Chomsky Normal Form (CNF) $S \rightarrow AB \mid aB$, $A \rightarrow aab \mid \varepsilon$, $B \rightarrow bbA$	*CO 2*	*PO 2*	**10**
		b)	Design a CFG for the language that generates the set of (i) All strings with exactly one a (ii) All strings with at least one a (iii) All strings with at least three a's Consider alphabet set as $\Sigma = \{a, b\}$.	*CO 2*	*PO 2*	**10**
			UNIT - IV			
	6	a)	i. Construct a Non-Deterministic Push Down Automata (NPDA) for accepting all palindromes over the alphabet set $\Sigma = \{a, b\}$ ii. Design Deterministic Push Down Automata (DPDA) for the language $L = \{ a^n b^{2n} \mid n >= 1\}$	*CO 3*	*PO 3*	**8**
		b)	Construct PDA that accepts the language generated by the grammar $S \rightarrow aSbb \mid abb$	*CO 3*	*PO 3*	**6**
		c)	Prove that the language $L = \{ a^n b^n c^n, n \geq 1 \}$ is not Context Free Language (CFL)	*CO 3*	*PO 3*	**6**
			UNIT - V			
	7	a)	Design a Turing Machine (TM) for set of string with equal number of 0's and 1's over $\{0, 1\}^*$	*CO 3*	*PO 3*	**10**
		b)	Determine whether the following (A, B) pairs have a Post Correspondence solution or not, if Yes give solution, if No why, Justify. (i) A= {b, babb, ba} B={bb, ba, ba} (ii) $A = \{1^2, 10^2, 1^3\}$ $B = \{1^3, 0^2 1, 1^2\}$	*CO 3*	*PO 3*	**10**

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